Ankle Joint / Complex

- It is a wt. bearing Joint.
- It permits both stability & mobility.
- Ankle complex consist of 28 bones and that forms 25 component jts.
  1) Proximal & distal tibiofibular jt.
  2) Talocrural jt
  3) Talocalcaneal jt / Subtalar jt
  4) Talonavicular jt
  5) Calcaneocuboid jt / transverse tarsal jt
  6) 5 tarsometatarsal jt
  7) 5 Metatarsophalangeal jt
  8) 9 Interphalangeal jt.
- The bone of the foot is divided into 3 functional Segments.
  1) Hindfoot (posterior segment)
     talus & Calcaneus.
  2) Mid foot (middle Segment)
     Navicular, Cuboid & 3 Cuneiform bones.
  3) Forefoot (anterior segment)
     Metatarsals and phalanges.

Motions -

- 3 motions
  dorsiflexion / Plantar flexion
  Inversion / Eversion

Abduction / Adduction

→ Dorsiflexion / plantarflexion -

- Occurs in sagittal plane & Coronal axis
- Dorsiflexion ↓es the angle b/w leg & dorsum of foot.
- Plantarflexion ↑es the angle b/w leg & dorsum of the foot.

At the level of toes.

Extension - bring the toes up

Flexion - bring the toes down

→ Inversion / Eversion -

- Occurs in the frontal plane & longitudinal axis
- Inversion - plantar surface of the segment is brought towards the midline.
- Eversion - plantar surface of the segment is away the midline.

→ Abduction / Adduction -

- Occurs in transverse plane & Vertical axis.
- Abduction - distal aspect of a segment moves away from the midline.
- Adduction - distal aspect of a segment moves towards the midline.

→ Pronation / Supination -

→ They are the components of Coupled motion.

- Pronation :- dorsiflexion, eversion, abduction.
- Supination :- plantarflexion, Inversion, adduction.

→ Varus / Valgus -

Valgus → ↑ in the medial angle b/w two bones.

Varus → ↓ in the medial angle b/w two bones.

Ankle Joint

→ It refers as "Talocrural jt"

→ Ankle is a Synovial hinge jt.

→ Consist of 1° of freedom

dorsiflexion / Plantar flexion.

Ankle jt Structure -

Proximal Articular Surface -

→ Proximal segment of ankle consist of Concave Surface.

→ Distal tibia and of the ~~d~~ tibial & fibular malleoli.

→ These three facets forms Concave jt surface.

→ Structure of the distal tibia and two malleoli resembles & reffered as "Mortise"

→ Adjustable mortise is more complex than a fixed mortise. Because it have mobility & stability functions.

Proximal Tibiofibular jt -

→ It is a "plane synovial jt"

→ Articulation - Head of the fibula

Posteriolateral aspect of tibia.

→ Common articulation pattern
Convex tibial facet
Concave fibular facet.
→ It is surrounded by a jt capsule that is reinforced by anterior & posterior tibiofibular ligament
→ Ankle plantar & dorsiflexion, it is a rotatory & translatory movements at proximal tibiofibular jt.

Distal tibiofibular jt -

→ It is a Syndesmosis or fibrous union.
→ Articulation - Concave facet of tibia
Convex facet of fibula.
→ Distal tibia and fibula are not in contact but are separated by fibroadipose tissue.

Ligaments -

→ Responsible for stable mortise.
→ Anterior & Posterior tibiofibular ligament
Interosseous membrane
→ Interosseous membrane directly suppots both proximal & distal tibiofibular articulations.
→ Function of talocrural jt is dependent on stability of the tibiofibular mortise.
→ The proximal tibiofibular jt must be mobile; if the proximal tibiofibular jt is mobile, so too must the distal tibiofibular jt be, because the two jts are mechanically linked.

Distal Articular Surface -

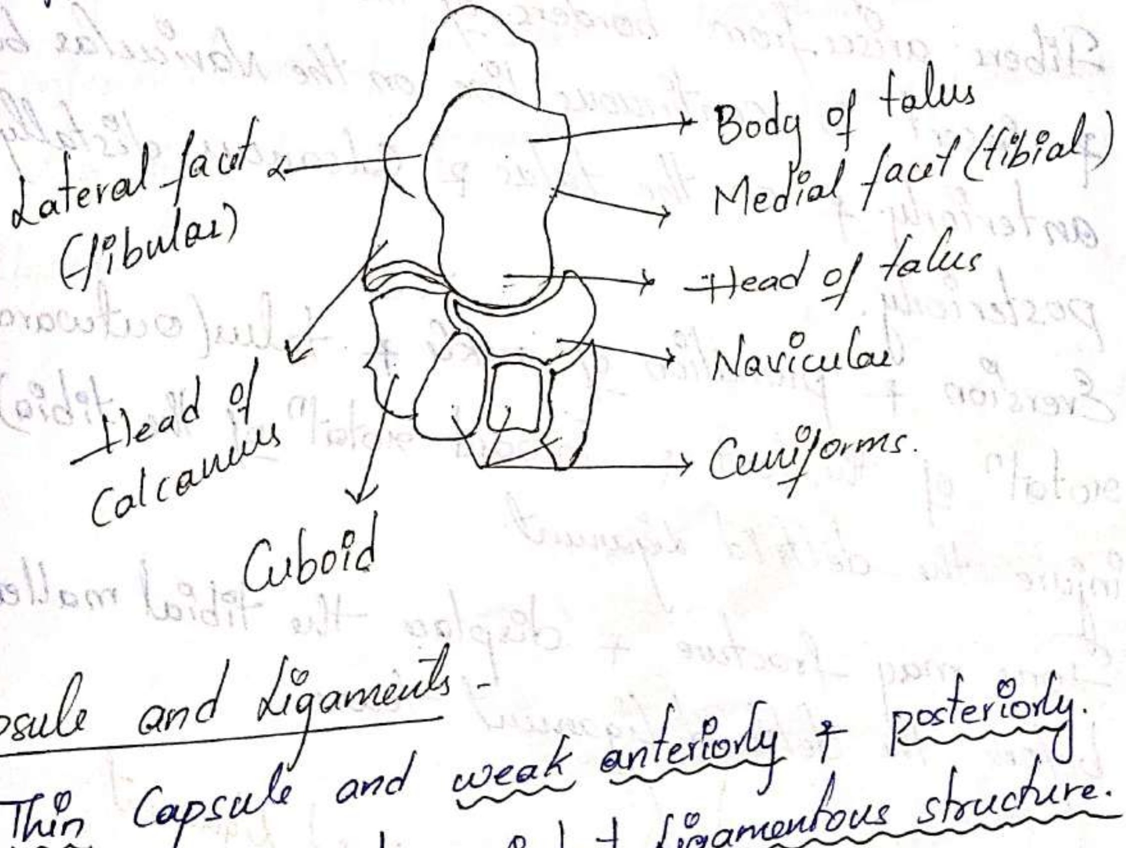
→ Distal articulation - Body of the talus.

→ Body of talus has 3 articular ~~sut~~ surfaces.

1) Large lateral (fibular) facet
2) Smaller medial (tibial) facet
3) Trochlear facet (superior).

→ It is a hinge variety of synovial jt.

→ The structural integrity of the ankle jt is maintained throughout the ROM of the jt by a no. of supportive structures.



Capsule and Ligaments -

→ Thin capsule and weak anteriorly & posteriorly.

→ Stability depends on intact ligamentous structure.

Ligaments -

Anterior & Posterior Tibiofibular ligament
Tibiofibular interosseous membrane
Medial collateral ligament
Lateral collateral ligament.

→ Anterior + posterior tibiofibular ligament and tibio-fibular interosseous membrane are imp for stability of the mortise + stability of the ankle.

→ Medial Collateral Ligament } Control medial - lateral
Lateral Collateral Ligament } stability.

Medial Collateral Ligament -

→ Also known as Deltoid Ligament.

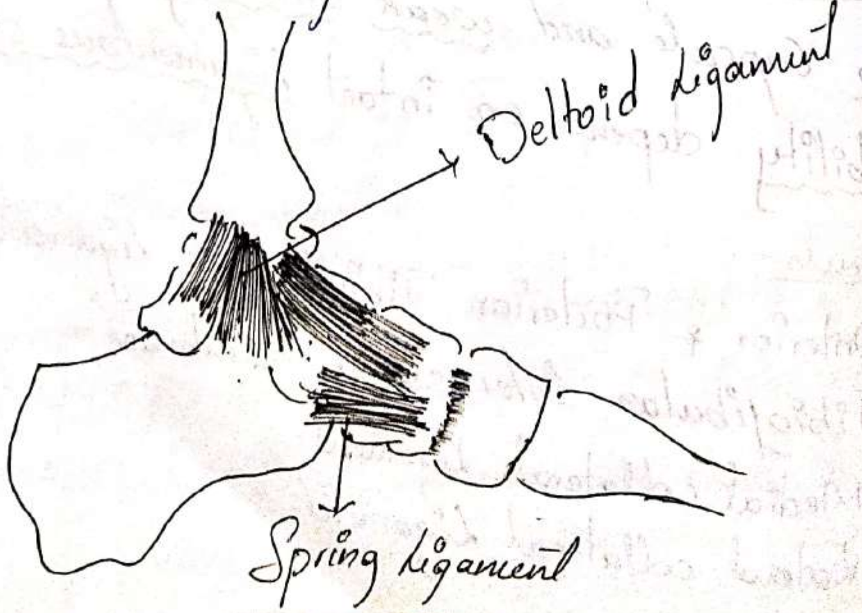
→ It is a fan-shaped.

→ It has superficial + deep fibers.

→ Fibers arises from borders of the tibial malleolus + insert in continuous line on the Navicular bone anteriorly + on the talus + Calcaneus distally + posteriorly.

→ Eversion + pronation of ankle + talus (outward rotatⁿ of the foot + inward rotatⁿ of the tibia) can injure the deltoid ligament.

→ Force may fracture + displace the tibial malleolus before the deltoid ligament tears.



Lateral Collateral Ligament -

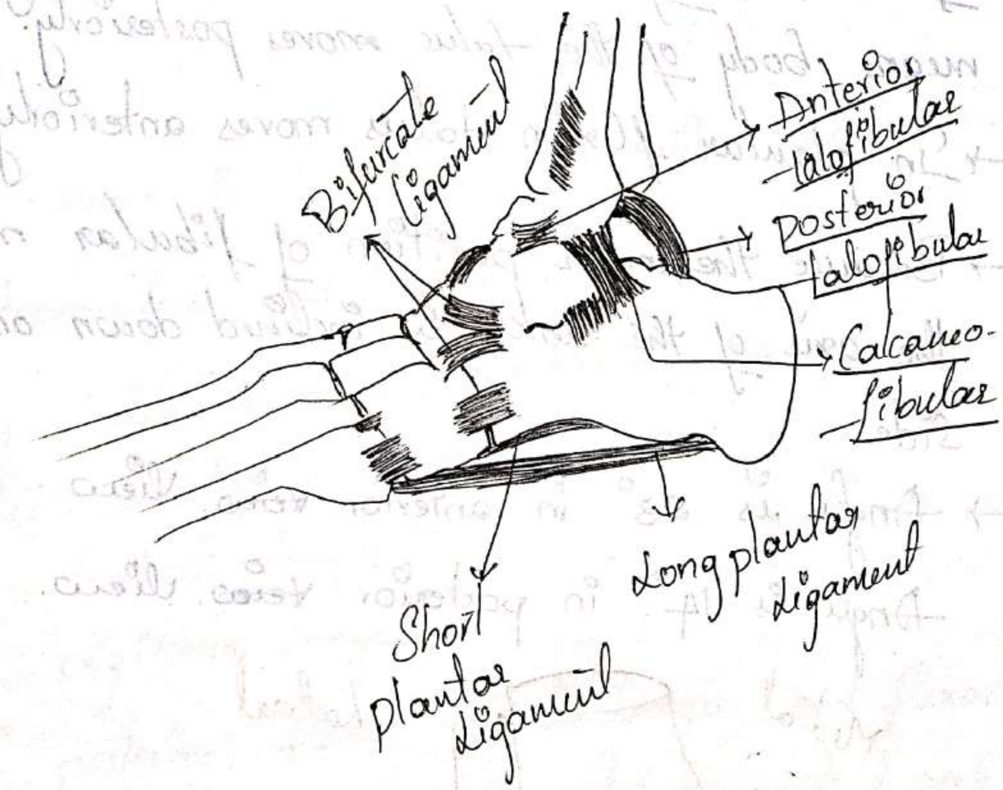
→ It Consist of 3 bands.
1) Anterior talofibular Ligament
2) Posterior talofibular Ligament
3) Calcaneofibular Ligament.

→ Anterior & posterior talofibular Ligaments run in horizontal position. (⟷)

→ Calcaneofibular Ligament is vertical. (↕)

→ Lateral collateral Ligament helps control inversion & supination of ankle and talus.

→ Inferior band of the superior peroneal retinaculum, which lies close & parallel to the calcaneofibular Ligament.



Axis -

→ The tibial torsion or tibiofibular torsion because both the tibia & fibula are involved with the rotation in transverse plane.

→ Tibial torsion accounts for the toe out position of the foot in the normal standing.

→ The torsion of the tibia is similar torsion found in the femoral shaft. Although normally reversed in direction.

→ The amount of outward tibial torsion ↑s from birth untill 10 yrs of age.

→ ~~As~~ tibial ~~por~~ torsion ↑s, the axis of the ankle jt is positioned more laterally in transver plane.

→ Classically the Ankle jt is considered to have 1° of freedom : Dorsiflexion / Plantar flexion.

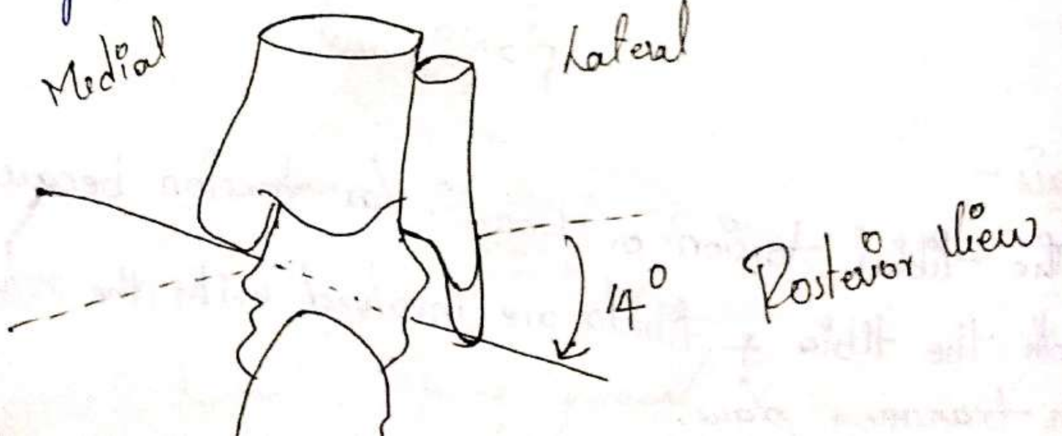
→ At the ankle the dorsiflexion of reports the motion of the head of the talus moves posteriorly. or upward means body of the talus moves posteriorly.

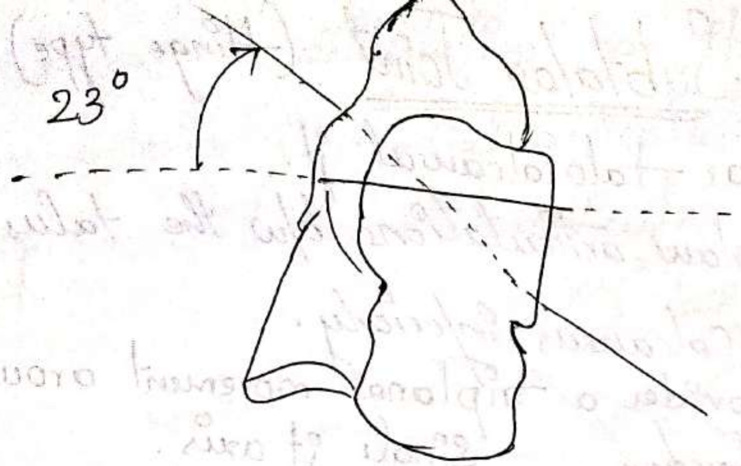
→ In plantar flexion talus moves anteriorly, downward

→ Because the lower position of fibular mallelous the axis of the ankle is inclined down on the lateral side.

→ Angle is 23° in anterior ~~veiw~~. View.

Angle is 14° in posterior ~~veiw~~. view.





Anterior View

Ankle Jt function -

→ 1° of freedom

ROM - dorsiflexion : 20°

plantarflexion : 50°

→ 10° of ankle dorsiflexion is considered the minimal amount needed to ambulate without deviation.

Muscles -

→ Triceps surae is the primary limitation to dorsiflexion (Gastronemius and Soleus)

→ Primary limitation to plantarflexion are

Tibialis anterior

Extensor hallucis longus

Extensor digitorum longus.

→ Tibialis posterior, flexor hallucis longus and flexor digitorum longus muscle help protect the medial aspect of ankle.

→ Peroneus longus & brevis muscles protect Lateral aspect.

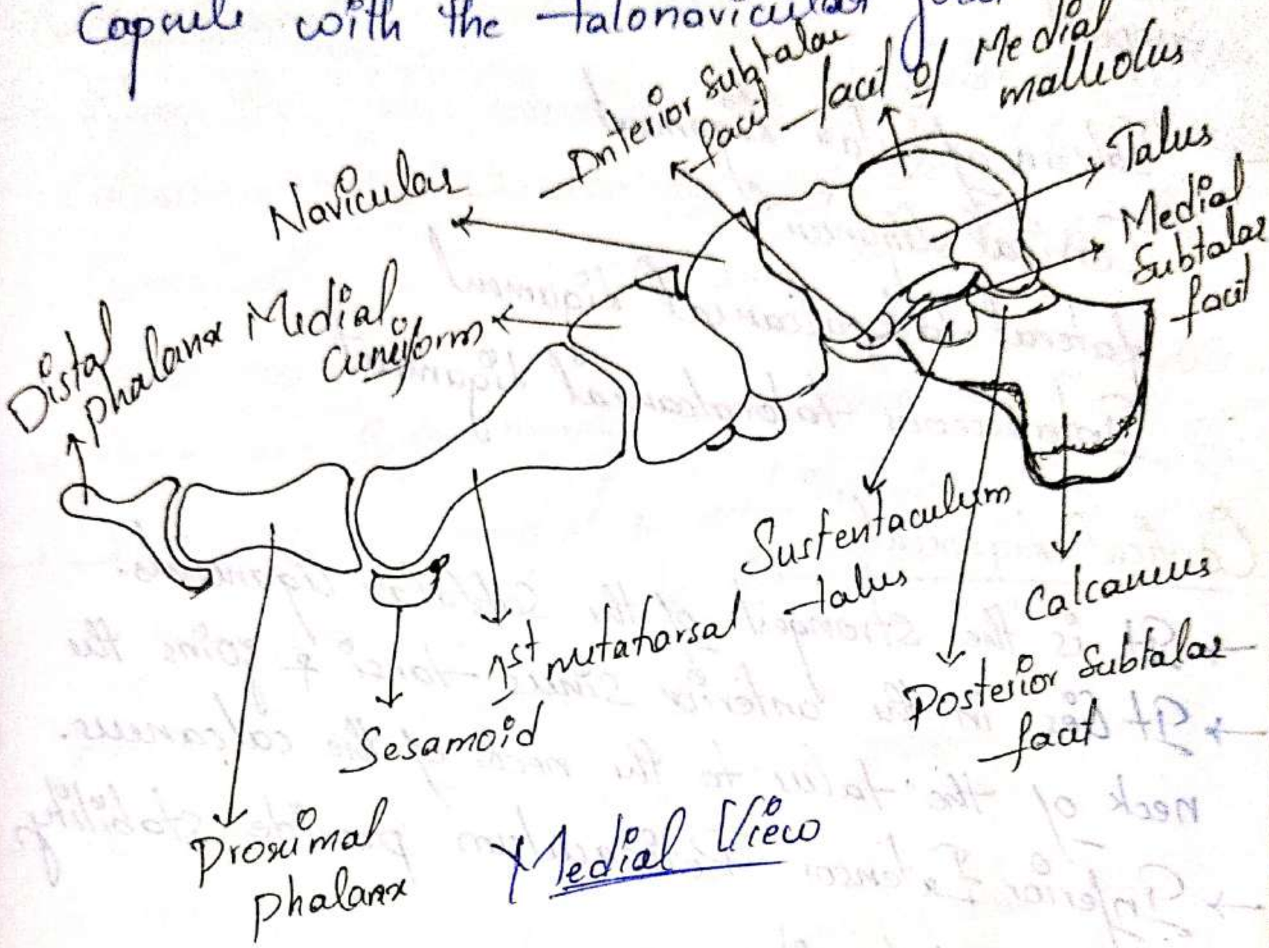
Subtalar Joint (Hinge type)

→ Also known as talocalcaneal jt.
→ 3 Separate plane articulations b/w the talus superiorly & calcaneus inferiorly.
→ 3 Surfaces provides a triplanar movement around a single jt axis.

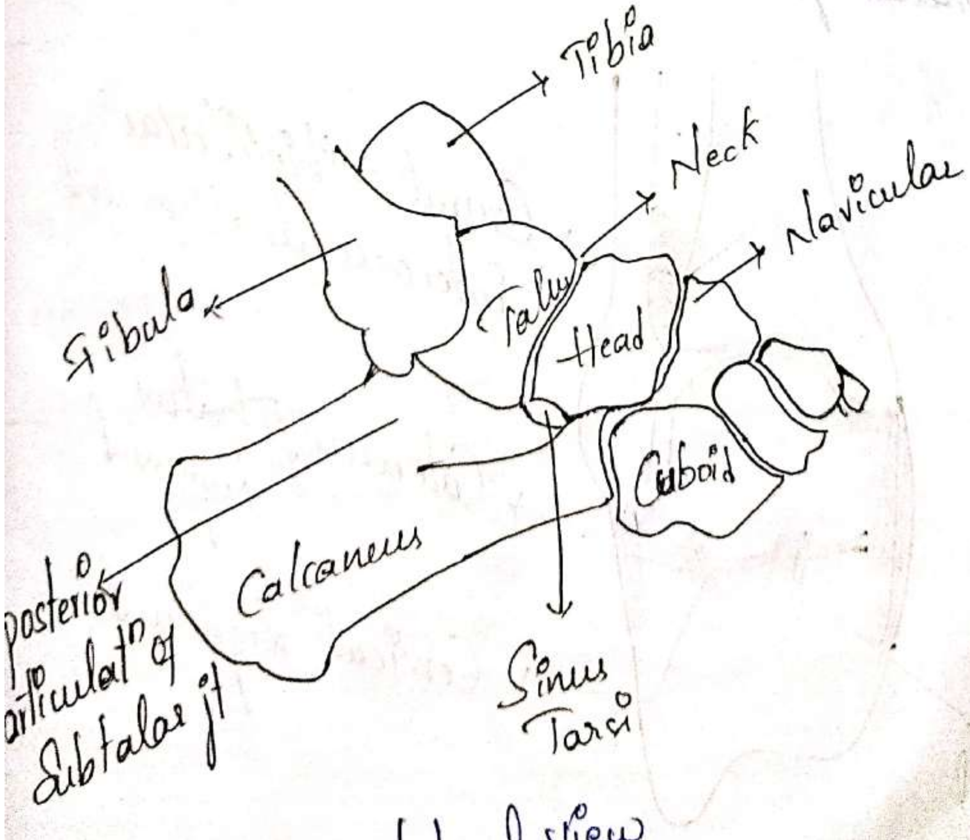
Subtalar jt Structure -

→ Posterior is the largest of 3 articulation.
→ Posterior articulation is formed by a concave facet on the talus & a convex facet on the body of the calcaneus.
→ Anterior and medial talocalcaneal articulations are ~~found~~ formed by a convex facet on the inferior body & neck of the talus and two concave facets on the calcaneus.
→ B/w posterior & anterior & medial articulations, there is a bony tunnel formed by a sulcus, in the inferior talus & superior calcaneus.
→ Funnel - shaped tunnel, is also known as tarsal canal, runs obliquely across the foot.
→ Anterior to the fibular mallelous there is a wide end of the tunnel, Called as Sinus tarsi.
→ Its smallest ends lies below the tibial malleoli & above a bony out cropping on the calcaneus, called as Sustentaculum tali.

→ The posterior articulation has its own capsule, the anterior & medial articulations share a capsule with the talonavicular joint.



Medial View



Lateral view

Ligaments -

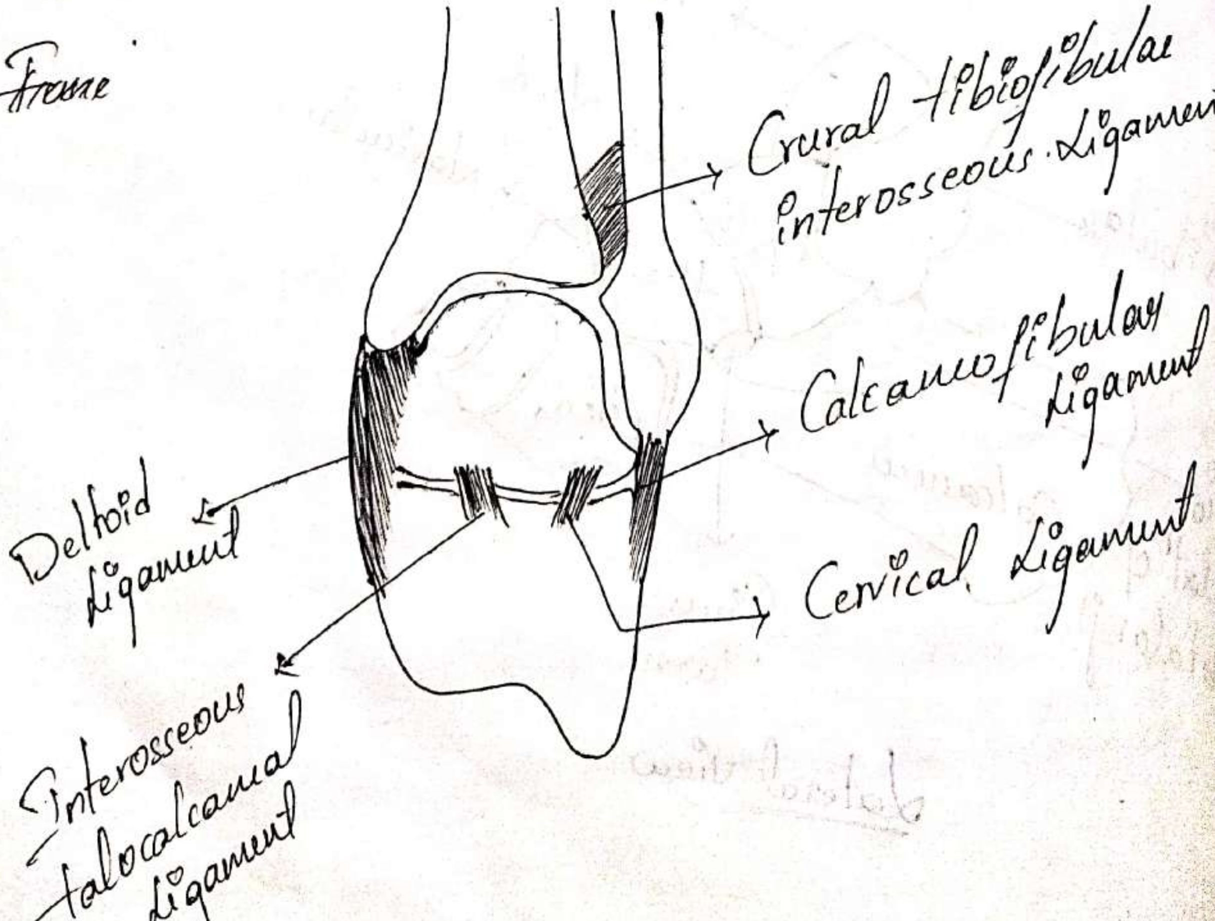
→ It is a stable jt, as well as strong ligamentous support.

→ Calcaneofibular Ligament
Cervical Ligament
Lateral talocalcaneal Ligament
Interosseous talocalcaneal Ligament.

Cervical Ligament -

→ It is the strongest of the subtalar ligaments.
→ It lies in the anterior sinus tarsi & joins the neck of the talus to the neck of the calcaneus.
→ Inferior Extensor Retinaculum provide stability to the subtalar jt.

Frame



Functions-

→ Pronation & Supination occurs in subtalar joint.

Osteokinematics

→ When the talus moves on the posterior facet of the Calcaneus, the articular surface of the talus should, theoretically, slide in the same direction as the bones moves.

→ A Concave surface moving on a stable convex surface.

→ However, at the medial & anterior joints, the talus surface should glide in a direction opposite to the movement of the bone.

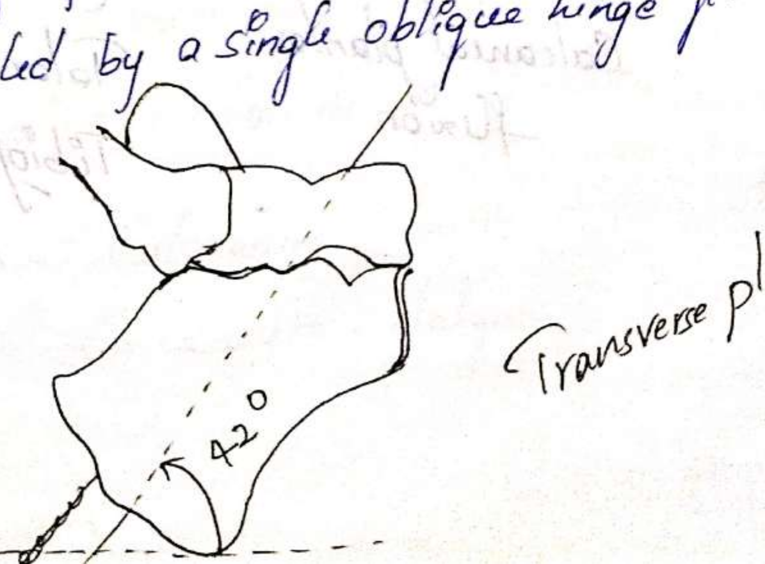
→ A convex surface moving on a stable concave surface.

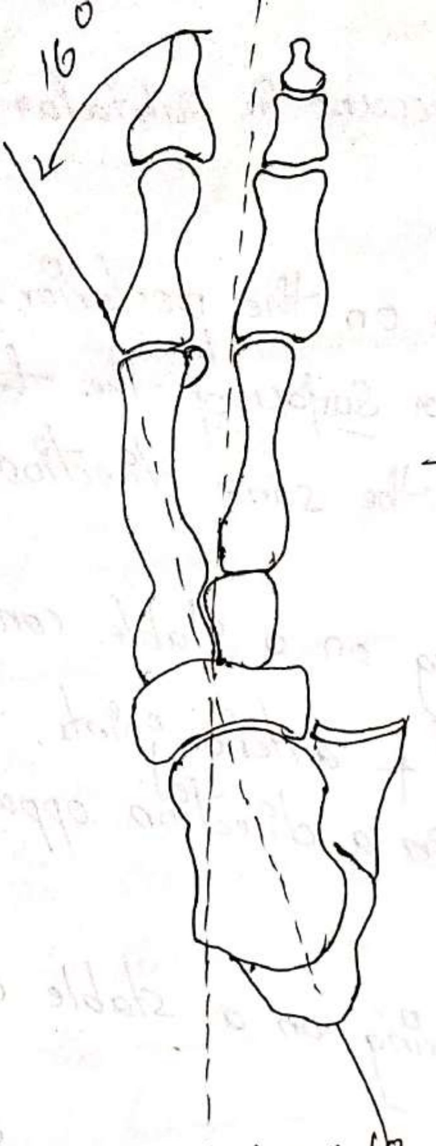
→ Pronation & Supination seen in oblique axis and transverse plane.

→ Average subtalar axis is inclined 42° upward & anteriorly from the transverse plane.

→ In the inclined medially 16° from the sagittal plane.

→ Supination & pronation (triplanar) ankle jt motion, can be modeled by a single oblique hinge jt.





Summary of coupled Motions :-

Coupled motion of Subtalar pronatⁿ / Supinatⁿ.

	Non-Wt. bearing	Wt. bearing
Supinatⁿ	Calcaneal inversion (or varus)	Calcaneal inversion (or varus)
	Calcaneal adduction	Talar abduction or (lateral rotation)
	Calcaneal plantar flexion	Talar dorsiflexion
		Tibiofibular lateral rotation.

	Non-Wt. bearing	Wt. bearing
pronat^{n}	Calcaneal eversion (or Valgus)	Calcaneal eversion (or valgus)
	Calcaneal abduction	Calcaneal adduct^{n} or (medial rotat^{n})
	Calcaneal dorsiflex^{n}	Talar plantarflexion
		Tibiofibular medial rotation.

Arthrokinematics -

→ Supination & pronations.

Non-Wt. bearing Subtalar jt motion -

Supination & pronation subtalar motion is describ-ed by its distal segment on the stationary talus on the lower leg.

Non-wt. bearing supinator is composed of the coupled calcaneal motions of abduction, inversion and plantar flexion.

Pronation of the non-wt. bearing calcaneal on the fixed talus & the lower leg is composed of the coupled motion of abduction, eversion and dorsiflexion.

Inversion - Calcaneum - Varus

Eversion - Calcaneum - Valgus.

Wt. bearing Subtalar jt motion -

When an individual is on wt. bearing the Calcaneus is on the ground & generally free to move around a longitudinal axis but limited in it ability to move around a coronal axis and vertical axis because of the Superimposed body wt.

Although wt. bearing calcaneus will continued to contribute the inversion & eversion component of the subtalar motions.

The other two component of coupled of the subtalar motion will be accomplished by the movement of the talus.

In wt-bearing supination the calcaneus continued to contribute the component of the inversion.

However calcaneus cannot adduct and plantar flexion these movements are achieved by the head of the talus.

Calcaneum - Inversion / Eversion

Talus - Adduction / Abduction

plantar flexed / Dorsi flexed.

Weight - Bearing Subtalar jt motion & its effect on the leg :-

→ During wt. bearing subtalar supinatn / pronatn, the coupled component motions of dorsiflexion / plantar flexn & abduction / adduction of the talar head require

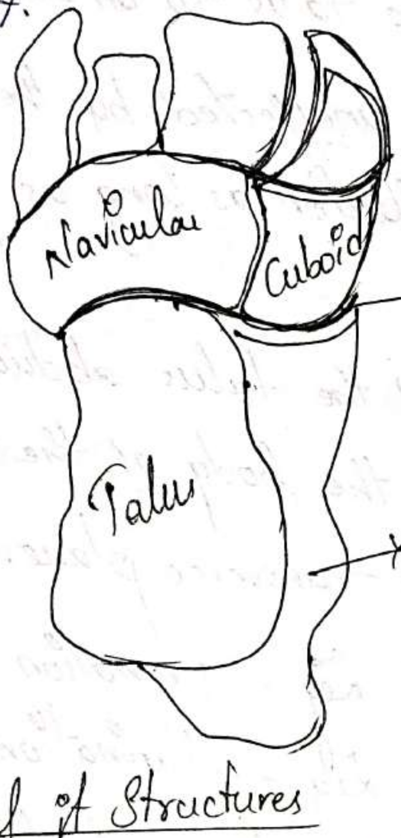
that the body of the move as well.
→ The body of the talus is of course, lodged within the superimposed mortise.
→ Dorsiflexion of the head of the talus required the body of the talus to slide posteriorly within the mortise.
→ Plantar flexion of the head of the talus requir-ed the body of the talus to anterior sliding.
→ The tibia remains uneffected by the talar dorsiflexion/ plantar flexion as long as the ankle jt is free to move.
→ When the head of the talus abduct in wt. bearing Subtalar supination the body of the talus must rotate laterally in transverse plane.
→ Medial rotation of Leg - pronation of foot
→ Lateral rotation of Leg - Supination of foot.

Subtalar Normal Position (Neutral) :-

→ Subtalar neutral position of Subtalar used it as a referance for a "normal" or "ideal foot" position.
→ The presumption that an individuals neutral Subtalar jt position may deviate from the point at which the midlines of the posterior calcaneus and the posterior leg coincide, with a medial ↑ in the angle referr-ed to as Valgus & an ↓ referend as the Varus.

Transverse Tarsal Joint

→ It is also known as Midtarsal or Chopart jt.
→ It is a Combination of talonavicular jt and Calcaneocuboid jt.
→ The 2 Joints together present an S-shaped jt.
→ Navicular & the Cuboid bones are immobile in the wt-bearing foot.



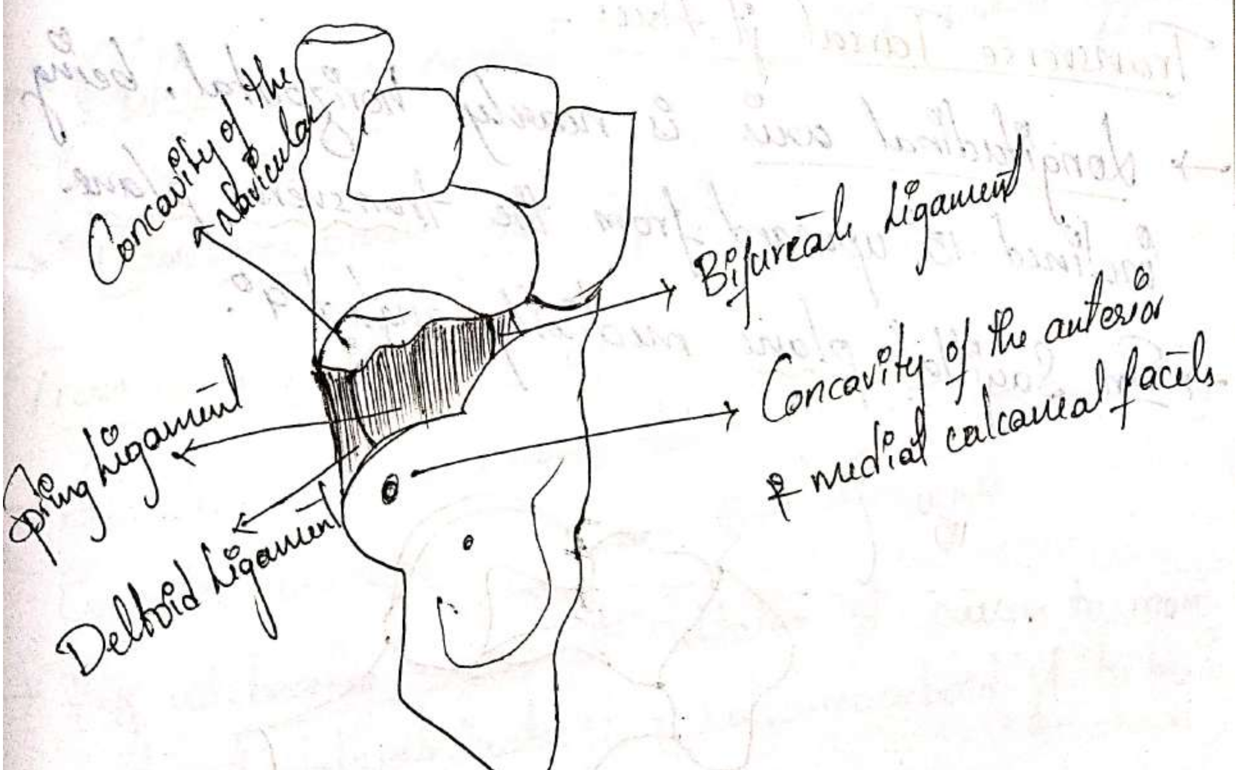
Transverse Tarsal jt Structures

Talonavicular jt - Ball & Socket jt

→ Proximal portion - Anterior portion of the head of the talus
distal portion - Concave posterior aspect of Navicular bone
→ Subtalar jt and transverse tarsal jt have a common jt capsule & inferior surface of this capsule is reinforced by "Spring Ligament" (plantar calcaneonavicular ligament)
→ Capsule is reinforced medially by deltoid & laterally by bifurcate Ligaments.

→ Spring ligament is a triangular sheet of ligamentous
→ Origin - from sustentaculum tali of calcaneum.
Insertⁿ - Inferior navicular bone.
→ Spring ligament is composed of 3 component
① Superomedial Band
② Medioplantar Band
③ Inferoplantar longitudinal Band.

→ Spring ligament plays an imp role
In supporting the head of the talus & talonavicular jt.
Functioning as one of the main static or passive
Stabilizers of medial longitudinal arch.



Calcaneocuboid jt -

→ Proximally - by anterior calcaneus
distally - posterior cuboid bone.

→ They having reciprocal shape, it makes available motion at the calcaneocuboid jt but it is a restricted type.

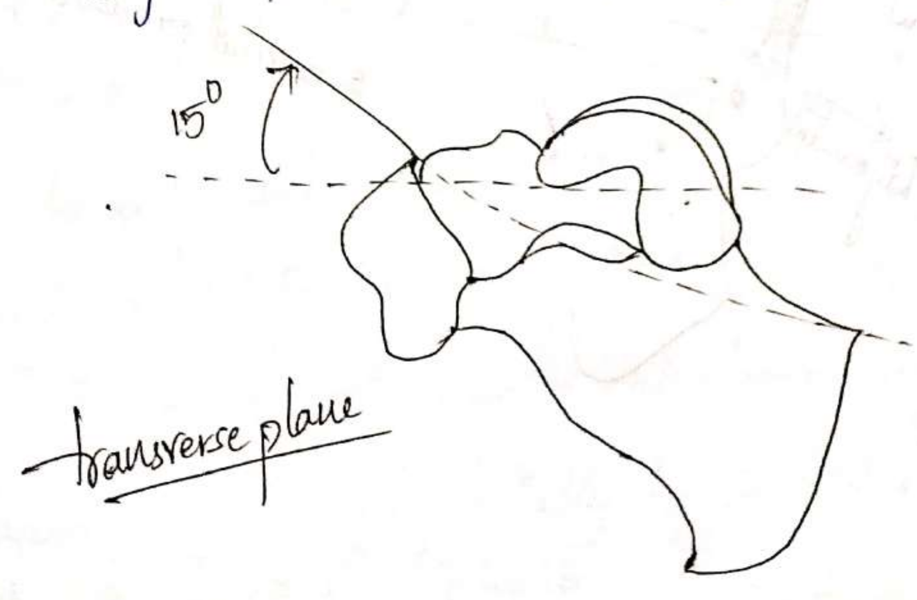
→ Calcaneocuboid have its own capsule.

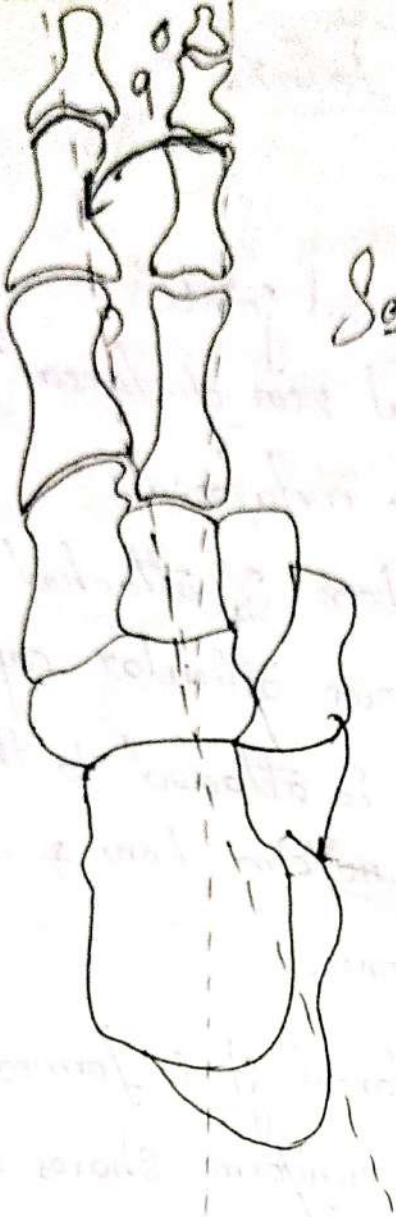
→ Capsule is reinforced laterally by the lateral band of the bifurcate ligament (calcaneocuboid ligament)
dorsally - dorsal calcaneocuboid ligament
Inferiorly - Plantar calcaneocuboid (short plantar) & long plantar ligaments.

Transverse Tarsal jt Axes -

→ Longitudinal axis is nearly horizontal, being inclined 15° upward from the transverse plane.

→ In Sagittal plane medially angled 9°.





→ Oblique axis medial to the Sagittal plane approximately 57°
→ Transverse plane 52° superiorly

Transverse Tarsal jt Function-
→ Subtalar and transverse tarsal jt are mechanically Connected because they share the jt capsule.
→ Any wt. bearing motion at Subtalar jt causes to movement at ~~talocalcaneal~~ jt + calcaneocuboid jt to begin to supinate.
→ It is a transitional like b/w hindfoot + forefoot.

Tarsometatarsal Joints :-

Structure -

→ It is a plane synovial joints.
→ Formed by the distal row of tarsal bones(posteriorly and the bases of the metatarsals.
→ The 1st metatarsal bone is attached to the medial Cuniform & has its own articular capsule.
→ 2nd metatarsal bone is attached to the mortise formed by middle cuniform bone & sides of medial & lateral cuniform bones.
→ The 3rd tarsometatarsal jt is formed by 3rd metatarsal & lateral cuniform, shares a capsule with Second tarsometatarsal jt.
→ The 4th & 5th tarsometatarsal jts are formed by the bases of the fourth & 5th metatarsals with Cuboid bone.
→ There is a deep transverse metatarsal ligament that Spans the heads of metatarsals on the plantar Surface.

Axis -

→ A "ray" is defined as a functional unit formed by metatarsal & its associate cuneiform bone.

→ Navicular & Cuboid bones are immovable so the ray is upto the cuneiform & metatarsal bones.

→ The 4^{th} & 5^{th} rays are formed by the metatarsals along because these are articulate with the cuboid bone.

→ 1^{st} & 5^{th} rays are most actively participated in the motion.

→ The jt axis which is passing is the oblique type jt axis.

→ Oblique axis passing through 1^{st} raye the motions are triplanar : Plantar & dorsiflexion

Inversion & Eversion

Abduction & Adduction.

* Dorsiflexion is accompained by eversion & abduction.
* Plantarflexion is accompained by Inversion & adductn.

→ The jt axis passing through 3^{rd} rays is coronal axis & the motion occuring are plantar & dorsiflexion.

Function -

→ It makes the hallowing / Flattening of the plantar surface.

→ In relation to the wt. bearing this joint contribute in regulation of the forefoot.

Supination twist -

→ When the hindfoot pronates substantially in wt. bearing the transverse tarsal jt generally will supinate to some degree to counter rotate the forefoot & keep the plantar aspect of the foot in contact with the ground.

→ If the range of transverse tarsal supination is not sufficient to meet the demands of pronating hindfoot, the medial forefoot will press into the ground & the lateral forefoot will tend to lift.

→ The 1st & 2nd rays will be push into dorsiflexion by the ground reaction force & the muscle controlling the 4th & 5th rays will plantarflex those tarsometatarsal jt in an attempt to maintain contact with ground.

→ Both dorsiflexion of 1st & 2nd rays of plantarflexion of 4th & 5th rays include component motion of the inversi of the rays.

→ Consequently the entire forefoot undergoes an inversion rotation around a hypothetical action at the second ray.

→ This rotation is referred to as the Supination twist

Pronation twist -

→ With hindfoot supination the forefoot tends to lift off the ground on its medial side & press into the ground on its lateral side.

→ The muscles controlling the 1^{st} & 2^{nd} rays will plantarflex those rays in order to maintain contact with the ground & 4^{th} & 5^{th} rays are forced into dorsiflexion by the ground reaction force.

→ Because eversion accompanies both plantarflexion of 1^{st} & 2^{nd} rays & dorsiflexion of 4^{th} & 5^{th} rays, the forefoot a whole undergoes a pronation twist.

→ These motions result in a pronation (Eversion) twist of the tarsometatarsal jts in an attempt to adjust the forefoot adequately.

Metatarsophalangeal Joints

→ The Metatarsophalangeal jts are Condyloid Synovial jt with 2° of freedom - Dorsi / plantar flexion
Abduction / Adduction.

Joint Structure-

→ MTP jts are formed proximally by convex heads of the metatarsals & distally by the Concave bases of proximal phalanges.

→ 1st metatarsal being longer than the 2nd & with other remaining progressively ↓ in length called Index plus.

→ 2nd metatarsal is being longer than the 1st, 3rd, 4th & 5th called Index minus / Morton's foot.

→ 1st metatarsophalangeal jt has 2 Sesamoid bones associated with it that are located on the plantar aspect of the 1st metatarsal head.

Sesamoidities

→ The Sesamoid bones & their supporting structures can become traumatized with excessive loading, leading to acute fracture.

→ Inflammatory conditions of the sesamoids which result in pain under the 1st metatarsal head are generally known as Sesamoidities.

→ Stability of metatarsophalangeal jts is provided by a joint capsule, collateral ligaments & plantar plates.

Joint function -

→ MTP jts have 2° degree of freedom.

→ Flexion / Extension is much greater than abduction / Adduction.

→ The MTP jts serves primarily to allow the wt. bearing foot to rotate over the toes through MTP Extension when rising on the toes or during walking

→ Limited extension ROM at the 1st MTP jt will interfere with the metatarsal break & known as Hallux rigidus.

Hammer Toe deformity -

→ Excessive extension at the metatarsophalangeal jt in a resting position is called a hammer toe deformity.

Interphalangeal Joints

→ IP jts of toes are synovial hinge jts with 1° of freedom : Flexion / Extension.

→ Greater toe has only one interphalangeal jt connecting 2 phalanges.

→ Four lesser toes have 2 interphalangeal jts (proximal & distal IP jts)

→ Toes function to smooth the wt. shift to the opposite foot in gait & help maintain stability by pressing against the ground in standing.

* Pes Cavus -

→ Pes Cavus / Cavovarus is described as high arched foot, both in wt. bearing + non. wt. bearing positions

* Hallux Valgus -

→ It is associated with a reduction in 1st MTP jt ROM.

→ Gradual Sublaxation of toe flexor tendons crossing the 1st MTP jt.

→ ↓ wt. bearing on great toe, & ↑ wt. bearing on 1st Metatarsal head.

* Flat Foot (Pes planus / pes valgus)

→ Defined as the progressive loss of dynamic & static medial longitudinal arch support.

→ The foot typically remains in a position of excessive pronation at the subtalar jt during wt. bearing.

Arches of Foot

Plantar Arches

→ It have 3 arches
1) Medial Longitudinal arch
2) Lateral Longitudinal arch
3) Transverse arch.

→ Medial Longitudinal arch is the largest arch.

Structure of the Arches -

→ Longitudinal arches are anchored posteriorly at the Calcaneus and anteriorly at the metatarsal heads.

→ Talus rests at the top of the foot and considered to be the "Keystone" of the arch.

→ Transverse arch is easily visualized in the midfoot at the level of tarsometatarsal jt.

→ Anterior tarsals, middle cuneiform bone forms the Keystone of the arch.

→ Shape and arrangement of the bones are partially responsible for stability of the plantar arches.

→ Inclination of the calcaneus and 1st metatarsal contribute to stability of medial longitudinal arch (standing)

→ Without support of Ligaments & muscles arches may collapse.

→ Support for medial longitudinal arch -
- plantar Calcaneonavicular ligament (Spring)
- Interosseous talocalcaneal ligament
- Deltoid Ligament
- Plantar aponeurosis

→ Spring Ligament supports the head of the talus & it is the most imp static stabilizer.

→ Long & Short plantar ligament support lateral longitudinal arch.

Function of the Arches -

→ Plantar arches contrasting both stability & mobility wt. bearing functions.

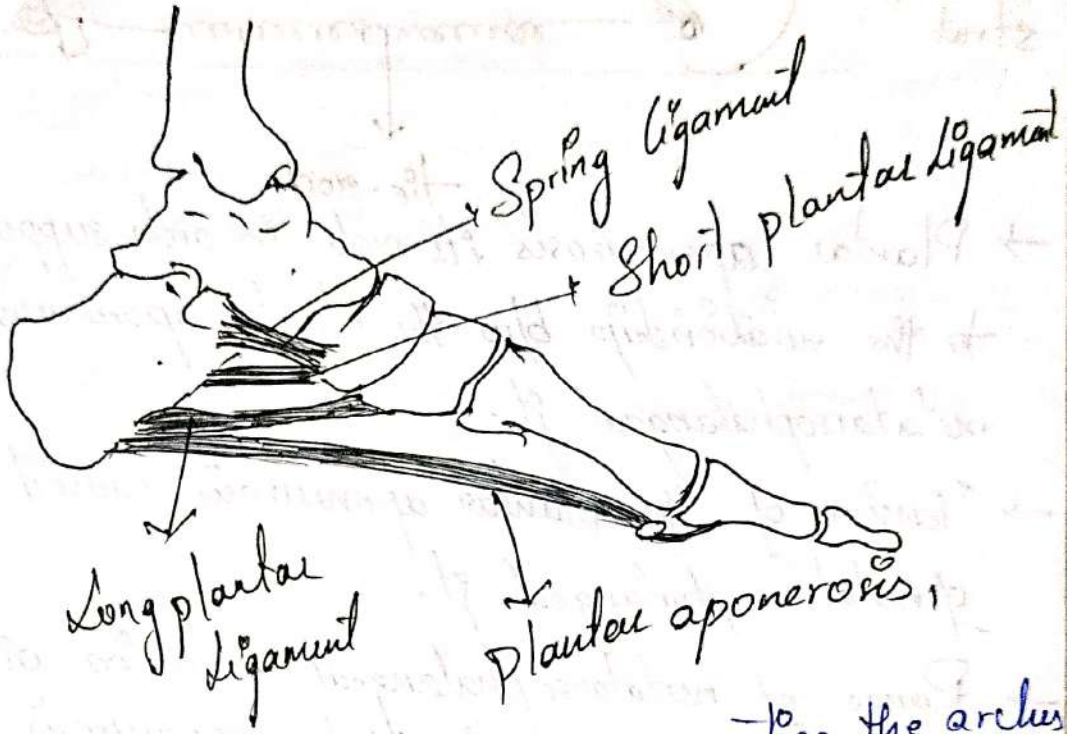
→ Wt. bearing Mobility function
1) dampen the impact of wt. bearing forces.
2) dampen superimposed rotational motions.
3) Adapt to changes in the supporting surface.

→ Wt. bearing Stability function
1) distribution of wt through the foot for proper wt. bearing.
2) Conversion of the flexible foot to a rigid level.

Plantar aponeurosis - or plantar fascia

→ It is a dense fascia.

→ Begins posteriorly on the medial tubercle of the calcaneus & continues anteriorly to attach plantar plates.



→ Function of the aponeurosis is supporting the arches to the function of a tie-rod on a truss.

→ Truss and tie-rod forms triangle.

→ Struts of the truss forms the side of the triangle and tie-rod to the bottom.

→ Calcaneum and talus forms one of the strut posteriorly.

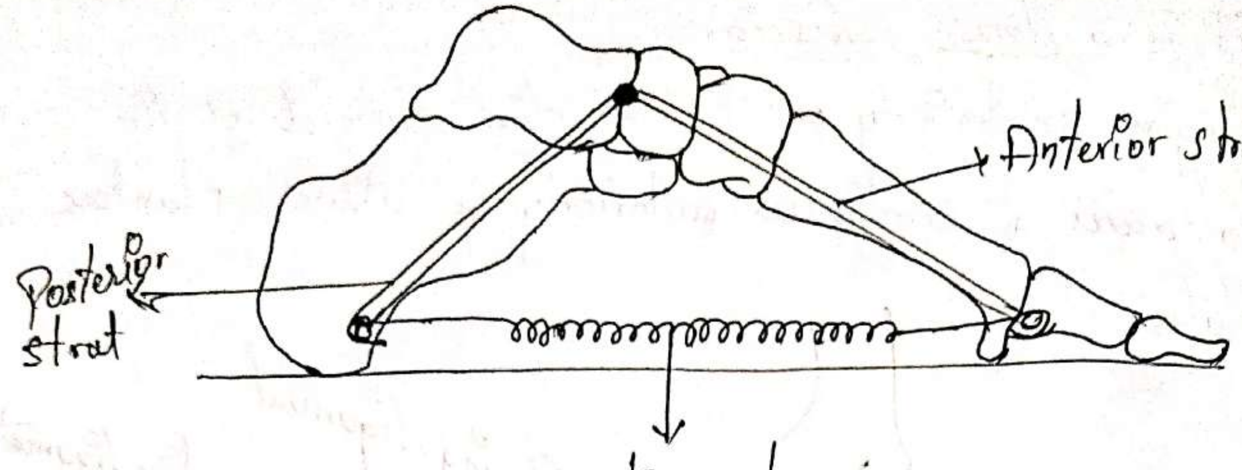
→ Anteriorly - tarsal & metatarsals.

→ Plantar aponeurosis act as tie-rod.

→ Wt. bearing foot -

struts (bones) are subjected to compression force.

tie-rod (aponeurosis) " " tension force.



→ Plantar aponeurosis its role in arch support are linked to the relationship b/w the plantar aponeurosis and the metatarsophalangeal jt.

→ Tension of the plantar aponeurosis caused by extension of metatarsophalangeal jt.

→ Range of metatarsophalangeal extension will be limited because further lengthening of the aponeurosis is limited.

Weight Distribution -

→ Bilateral stance -
  - Talus receives 50% of wt. bearing. (body. wt.)

→ Unilateral Stance -
  - Talus receives 100% of wt or body wt.

→ Standing position
  - Talus receives 50% of wt through larger posterior

Subtalar to the calcaneus.

→ 50% or less wt anteriorly through talonavicular and calcaneocuboid.

plantar pressure → are much greater during walking than during standing.

→ Factors include physical characteristics, age, body wt, height, foot structure → arch height, hallux length, joint motion.
Gait style and muscle action.

→ Excessive plantar pressures leads to pain and injury.

→ Structural & functional factors.
Hammer toe deformity, soft tissue thickness, Hallux ~~Lu~~ Valgus, foot type, walking speed

Muscles-

→ Dynamic Arch support - Extrinsic muscles -
Tibialis Posterior & Anterior
Peroneus longus & tertius
Flexor digitorum longus.

→ Medial arch longitudinal support - Tibialis posterior.

→ Intrinsic Muscles -
Abductor hallucis
flexor hallucis brevis
Flexor digitorum brevis
abductor digit minimi
dorsal interossei